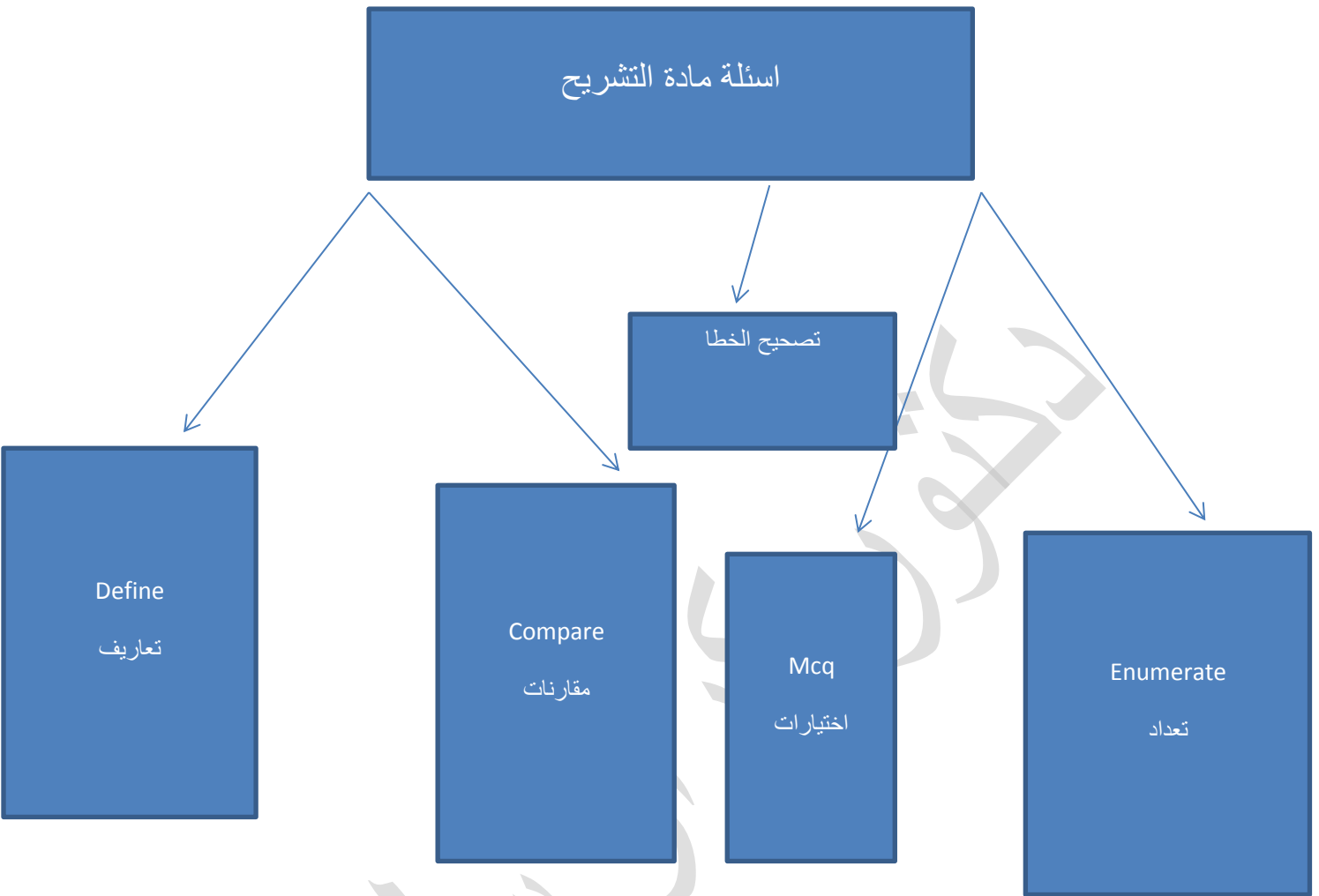


Dental Anatomy

بخصوص اسئلة التشريح وكيفية يتم دراسة المادة يكون كالآتي:



Dental Anatomy

MCq

Tooth	eruption
1-permanent maxillary central incisor	7-8 years
2-permanent lateral incisors	8-9 years
3-permanent mandibular incisors	6-7 years
4-permanent maxillary canine	11-12 years
5-permanent mandibular canine	9-10 years

Tooth	Eruption
6-permanent lateral incisor	8-9 years
7-maxillary 1 st premolar	10-11 years
8-maxillary 2 nd premolar	10-12 years
9-mandibular 1 st premolar	10-12 years
10-mandibular 2 nd premolar	11-12 years

Tooth	Eruption
Maxillary first molar	6-7 years

Tooth	Eruption(in months)
1-mandibular central incisors	6.5
2-maxillary central incisors	7
3-maxillary lateral incisors	7.5
4-mandibular lateral incisors	8
5-maxillary first molar	12-16

Tooth	Eruption(in months)
1-mandibular first molar	12-16
2-maxillary canine	16-20
3-mandibular canine	16-20
4-mandibular second molar	20-30
5-maxillary second molars	20-30

Dental Anatomy

Measurements in (mm)

Tooth	Length of the crown	Length of the root
1-permanent maxillary central incisor	10.5	13.0
2-permanent lateral incisor	9.0	13.0
3-permanent mandibular incisor	9.5	12.5
4-permanent maxillary canine	10.0	17.0
5-permanent mandibular canine	10.5	13.0
6-permanent mandibular lateral incisor	9.5	14.0

Tooth	Length of the crown	Length of the root
1-maxillary first premolar	8.5	14.0
2-maxillary second premolar	8.5	14.0
3-mandibular first premolar	8.5	14.0
4-mandibular second premolar	8.0	14.5
5-maxillary first molar	7.5	B-12 L-13

Maxillary central incisor

Note: ALL incisors develop from four lobes three labial lobes and one lingual lobe (mcq)

1-Almost straight mesial outline and rounded distal outline (mcq)

2-sharp mesio-incisal angle and rounded disto-incisal angle

3-single tapered root

mcq

النقاط ١ و ٢ و ٣ خصائص عامة لل incisor

Labial Aspect	Distal Aspect	Incisal Aspect
The distal outline is more convex than the mesial outline with the crest of curvature junction between incisal and middle thirds	There is little differences between distal and mesial outline The curvature of cervical line is less distally than mesially	The mesio-distal dimension labially is greater than that lingually
✕	✕	✕

Maxillary Lateral Incisor

Principle identifying features?(mcq)

1-the crown is more rounded,shorter and narrower mesio-distally	2-the mesio-incisal angle is acute and disto-incisal angle is more rounded
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Aspects

Labial aspect	Mesial aspect	Distal aspect	Lingual aspect	Incisal aspect
1-height of mesial contour junction middle and incisal third	1-mesial contact area junction of incisal and middle third	1-the curvature of cervical line is less than that mesially	لا توجد اي امسكيو لحد الان فقط مجرد اطلاع عليـة	The incisal aspect resembles that of the maxillary central incisor but
2-height of distal contour is center of the middle third	2-the crown narrowing labio-lingually	2-it not uncommon to find a developmental groove extending to the root	✖	A-the cingulum and incisal ridge may be larger
✖	✖	3-there is developmental groove and depression in this aspect	✖	B-the labio-lingual to mesio-distal measurement is larger

Q:numerate the variations from typical form(Anomalies)?

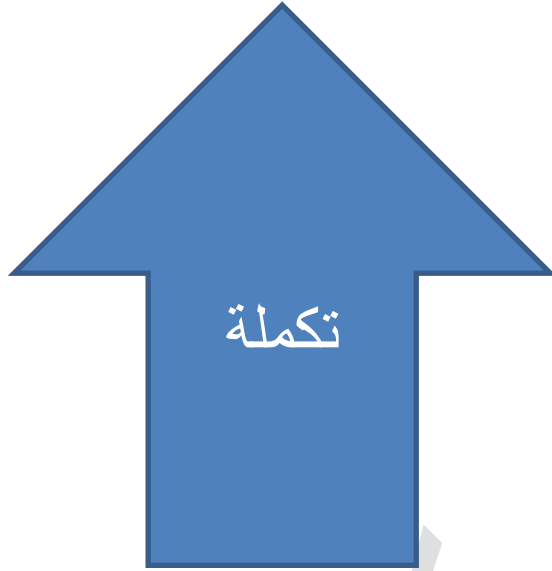
1--peg-shape lateral incisor with a thin root and small	2- the maxillary lateral incisor may be congenitally	3- large developmental groove on the distal aspect
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Dental Anatomy

conical crown

missing

extending to the root



Permanent Mandibular incisors

Characteristic features of permanent mandibular incisors?(mcq)?

1-the mandibular incisors have smaller mesio-distal dimension

2-the contact areas near the incisal ridges mesially and distally

ملاحظة: يجب دراسة ال الاسبيكت في الملزمة

حصرا

Permanent mandibular lateral incisors

The mesial side of the crown is longer than the distal side



Dental Anatomy

Common characteristics of permanent Canines?(mcq)

1-they are develop from four lobe 3 labial and 1 lingual	2-they are wider buccolingually than mesiodistally	3-their distal cusp slope is longer than the mesial cusp slope
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The permanent Maxillary canine

Principle identifying features?

1-very long single root	2-the distal slope of the cusp is longer than the mesial slope	3-single pointed cusp
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Aspects

Labial aspect	Lingual aspect	Distal aspect	Incisal aspect
Mesial out line—curvature junction between incisal and middle third	Two fossa	The curvature of cervical line is less distally than mesially	1-the geometric shape is diamond
	Cingulum larger	The developmental depression of the root is more pronounced distally than mesially	2-the labio-lingual dimension is greater than the mesio-distal dimension
	There is developmental depression mesially and distally extending for most of the root length		

The permanent Mandibular canine

Principle identifying features? (mcq)

1-the mesial slope of the cusp is shorter than the distal slope

2-the mandibular canine narrower mesio-distally and labio-lingually than the maxillary canine(smaller)

Aspect

Incisal aspect

Geometric shape: Oval

Dental Anatomy

Q1; Compare Between upper and lower Canine?

Upper canine	Lower canine
1- wider	1- narrower mesio-distally than labio-lingually
2- single pointed cusp	2- the cusp not well develop
3- the tip of the ucps is aline with the center of root	3- the tip of the cusp is not aline with the center of the root
4- distal contac area is located of middle third	4- the disal contact area is located more incisally than upper canine
5- the lingual surface of the crown is longer	5- the lingual surface is shorter
6- lingual ridge develop	6- lingual ridge less develop
7- mesial contact area junction of middle and incisal third	7- mesial contact area near mesio-incisal angle

Q2; compare Between **maxillary** central incisor and lateral incisor?

Central and lateral incisors	
	Maxillary central incisor
1. Labial Aspect	1. The crown is longer and wider mesio-distally. 2. Mesial outline is almost straight with slight convexity with a crest of curvature at the mesio-incisal angle. 4. Distal outline is more convex than mesial outline with the crest of curvature at the junction between the incisal and the middle thirds. 5. The root is shorter with a cone-shaped and blunt apex.
2. Mesial Aspect	5. The crown is wider labio-lingually with a pronounced curvature of the cervical line apically.
3. Distal Aspect	6. No developmental groove extending to the root.
4. Lingual Aspect	7. Well defined lingual anatomical features.
	Maxillary lateral incisor
1. Labial Aspect	1. The crown is shorter and narrower. 2. Mesial outline is similar to maxillary central incisor with acute and more rounded mesio-incisal angle with the crest of curvature between middle and incisal thirds. 3. Distal outline is more rounded and the crest of curvature is at the center of the middle third. 4. The root is longer than maxillary central incisor, tapers evenly and curves distally in a pointed apex.
2. Mesial Aspect	5. The crown is narrower labio-lingually with the curvature of the cervical line less pronounced.
3. Distal Aspect	6. A developmental groove extending from the side of the cingulum to the root may be found.
4. Lingual Aspect	7. More prominent cingulum with more concave lingual fossa.

Dental Anatomy

Q3:compare Between Maxillary and Mandibular Central Incisor?

	upper teeth	lower teeth
Feature	Max. central incisor	Mand. central incisor
Location of the incisal ridge	Centered over the root.	More lingually positioned.
Labial outline (from mesial aspect)	Convex cervically, slightly convex to the incisal ridge.	Slightly convex cervically, straight to the incisal ridge.
Lingual aspect	Feature full.	Featureless.
Distal Contact area	At the junction of incisal and middle thirds.	More incisally to the junction of incisal and middle thirds.
Size	Larger.	Smaller.

ملاحظة: ممكن ان ياتي من المقارنات امسكيو او تصحيح الخطا ارجوووووووووووووووووو الانتباه على هذا الشئ المهم جدا

ملاحظة: بخصوص الاسيكت بعض الجهات لم يتم ذكرها وبعض الجهات تم ذكر نقاط قليلة منها ان هذه النقاط التي تم ذكرها هي تعتبر نقاط مهمة جدا واتت في الامتحانات لذلك يجب التركيز عليها اما التي لم يتم ذكرها فيجب مراجعتها

Each cusp composed :	4Ridges
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Upper jaw=Maxillary	fixed
Lower jaw=Mandibular	movable

Anterior teeth	Incisors and canines
Posterior teeth	Premolars and molars

Deciduous teeth are 20 in number 10 in each arch and 5 in each quadrant	Formula: $I \frac{2}{2} C \frac{1}{1} M \frac{2}{2} = 10$ (for each side)
permanent teeth 32 in number 16 in each arch and 8 in each quadrant	$I \frac{2}{2} C \frac{1}{1} P \frac{2}{2} M \frac{3}{3} = 16$ (for each side) I =incisor C =canine P =premolar M =molars



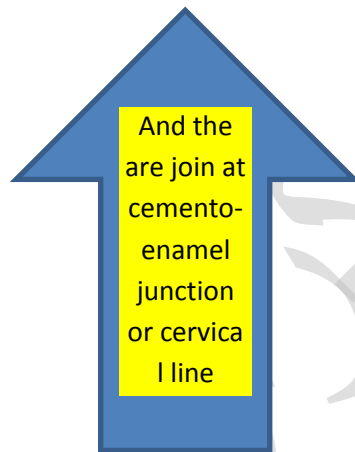
Dental Anatomy

All teeth have their mesial surface touching the distal surfaces of the adjacent tooth except the	Maxillary and mandibular central incisors both permanent and deciduous
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Line angle	Junction two surfaces
Point angle	Junction of 3 surfaces

Each tooth has	Crown and Root
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The crown is covered with	enamel
The root covered with	cementum



Dentin+enamel+cementum=	Hard tissue of the tooth
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Dental pulp=	Soft tissue of the tooth and present in A-pulp chamber; is the part of dental pulp in the crown B-pulp canal; is the part of dental pulp in the root
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The number of roots?

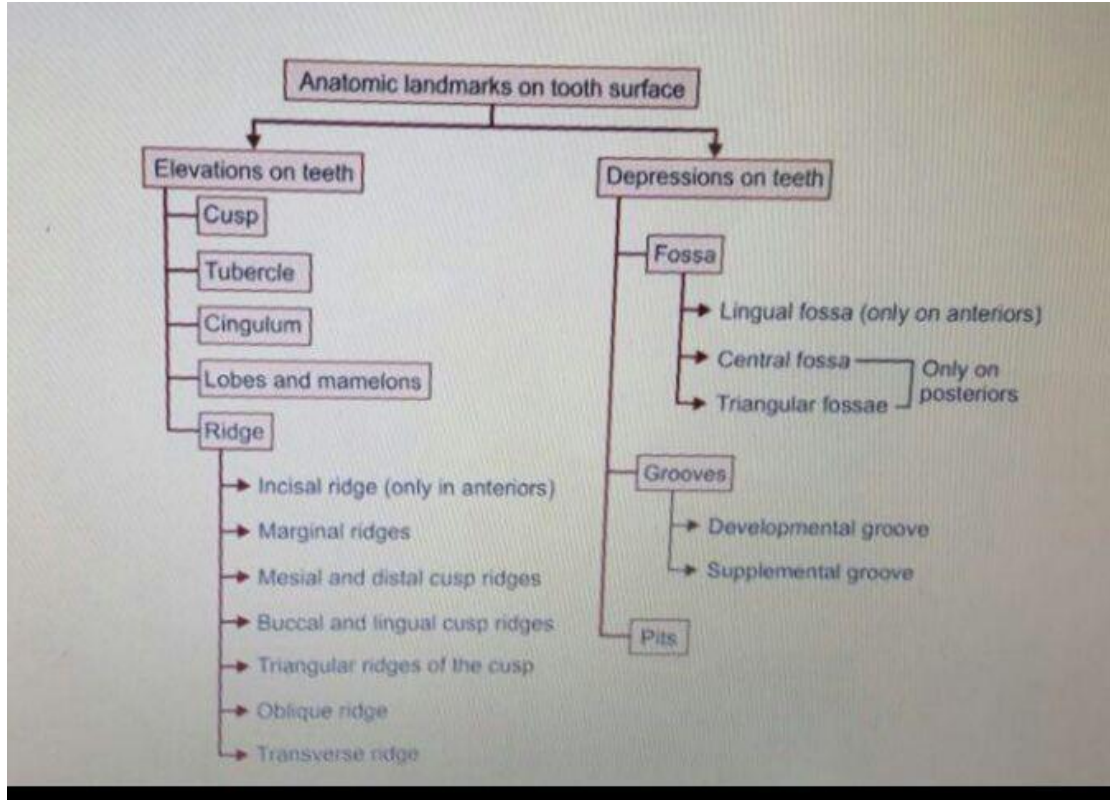
Single root; in all anterior teeth ,lower premolars upper second premolar	Two root; in lower molars, upper first premolar	Three root; in upper molars
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ملاحظة: انظمة تسمية الاسنان تدرس في الملزمه حصرا ان هذا الموضوع مهم جدا وفي الغالب ياتي عليه مابين

1-2 Marks

Dental Anatomy

Anatomic landmarks on tooth surfaces



MCQ+define

Q; define the cusp? And numerate the ridges that is present on the buccal cusp of the mandibular 1st premolar (١٤٤٣هـ)?

Cusp; is an elevation on the crown portion of a tooth making up a divisional part of the occlusal surface. Cusps are present in posterior teeth and canines

1-Buccal and distal cusp ridges	2-Mesial and distal cusp ridges	3-Triangular ridge of the cusp
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Q;Each cusp is gothic pyramid with four sides four ridges?enumrate this ridges

1-Mesial and distal cusp ridges	2-Buccal/Lingual cusp ridges	3-Triangular ridge of the cusp
Mcq; its enamel abnormality	tubercle	✗

Dental Anatomy

Premolars

Mandibular first premolar	Mandibular second premolar
1-number of root;single root	1-number of root;single root
2-shape of occlusal; circular	2-shape of occlusal;square(2cusp) Or 3cusp(rounded)
3-size; smaller	3-size; Larger
4-number of cusp;two cusp buccal,lingual cusp	4-A-two cusp 1 buccal,1lingual B-3cusp 1buccal,2lingual
5-non functional lingual cusp	5-functional lingual cusp

Maxillary first premolar	Maxillary second premolar
1-number of root; two root buccal,lingual	1-number of root;single root
2-number of cusp;two cusp sharply buccal longer than lingual cusp	2-number of cusp;buccal and lingual cusp are equal in height
3-shape of occlusal;hexagon	3-shape of occlusal; oval
4-supplemental; No	4-supplememntal;with more
5-Developmental depression;on the mesial surface of the crown	5-NO
6-central D.G=interrupts the mesial-marginal ridge	6-C.D.G=Shorter

Q;enumerate the fossa are present in the maxillary 1st premolar?

Mesial triangular fossa

Q;enumerate the pits found in the occlusal of the maxillary 1st premolar?

1-mesial-developmental groove	2-Distal developmental groove
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Q;enumerate the groove are found in the occlusal aspect of maxillary 1st premolar?

Dental Anatomy

1-mesio-buccal groove	2-disto-buccal groove
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Q;compare between 3cusp type and 2 cusp type of the lower 2nd premolar?

3 cusp	2 cusp
1-shape; square	1-shape; rounded
2-number of cusp; 3 buccal cusp is largest	2-number of cusp; there is one well develop cusp
3- supplemental groove; are often seen	3- NO

Q;enumerate the fossa are found in the lower 1st premolar?

1-mesial triangular fossa	2- distal triangular fossa
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Q;enumerate the groove in the occlusal of lower 2nd premolar?

1-mesial developmental groove	2-distal developmental groove	3-lingual developmental groove
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slopes

1-upper first premolar	2-upper second premolar	3-lower first premolar	4-lower second premolar	5-upper first molar
Mesial slope of the buccal cusp is longer than the distal slope	The mesial slope of the buccal cusp is shorter than the distal slope	The mesial slop of the buccal cusp is shorter than the distal slope	The mesial slop of the buccal cusp is shorter than the distal slope	Mesiobuccal cusp is broader than the distobuccal cusp

Lower molars

Compare between upper 1st molars and lower 1st molars?

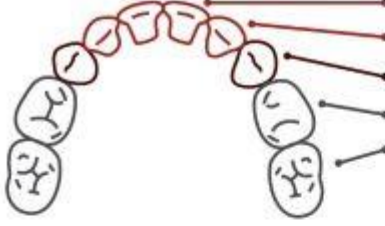
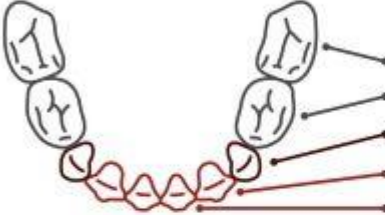
Upper first molar	Lower first molar
4 cusp	5 cusp
Oblique ridge; yes	No
rhomboidal	rectangular
3 roots	2 roots
Greater bucco-lingually	Greater mesio-distally

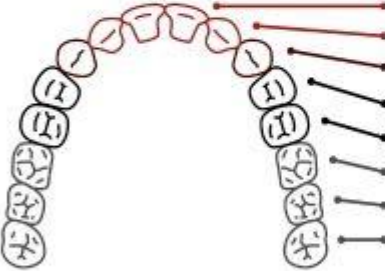
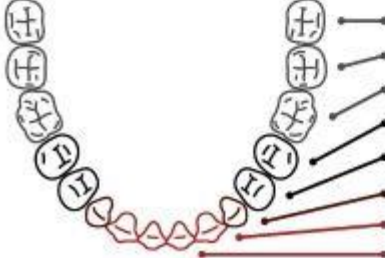
Proximal contacts(contact area)

Maxillary	Mesial	Distal
1-Central incisor	Incisal 1/3	Junction of incisal and middle 1/3
2-Lateral incisor	Junction of incisal and middle 1/3	Middle 1/3
3-Canine	Junction of incisal and middle 1/3	Middle 1/3
4-1 st pre	Middle 1/3	Middle 1/3
5-2 nd pre	Middle 1/3	Middle 1/3
6-1 st molar	Middle 1/3	Middle 1/3
7-3 rd molar	Middle 1/3	n/a

Maxillary	Mesial	Distal
8- 2 nd molar	Middle 1/3	Middle 1/3

Dental Anatomy

BABY (DECIDUOUS) TEETH CHART		UPPER TEETH	TOOTH ERUPTS	TOOTH LOST
	Central incisor		8-12 months	6-7 years
	Lateral incisor		9-13 months	7-8 years
	Canine		16-22 months	10-12 years
	1st Molar		13-19 months	9-11 years
	2nd Molar		25-33 months	10-12 years
	2nd Molar	LOWER TEETH	23-31 months	10-12 years
	1st Molar		14-18 months	9-11 years
	Canine		17-23 months	9-12 years
	Lateral incisor		10-16 months	7-8 years
	Central incisor		6-10 months	6-7 years

ADULT (PERMANENT) TEETH CHART		UPPER TEETH	TOOTH ERUPTS
	Central incisor		7-8 years
	Lateral incisor		8-9 years
	Canine		11-12 years
	1st Premolar		10-11 years
	2nd Premolar		10-12 years
	1st Molar		6-7 years
	2nd Molar		12-13 years
	3rd Molar	LOWER TEETH	17-21 years
	2nd Molar		11-13 years
	1st Molar		6-7 years
	2nd Premolar		11-12 years
	1st Premolar		10-12 years
	Canine		9-10 years
	Lateral incisor		7-8 years
	Central incisor		6-7 years

Dental Anatomy

Outline	Mesial	Distal
Mandibular Central	Straight Tapers evenly from the mesioincisal angle to the cervix	Straight Tapers evenly from the mesioincisal angle to the cervix
Mandibular Lateral	Straight Longer than Distal outline	Slightly convex
Mandibular Canine	Nearly straight with mesial outline of the root	Slightly convex
Mandibular 1st Premolar	Straight or slightly concave above the cervical line to a point where it joins the curvature of the mesial contact area	Slightly concave above the cervical line to a point where it is confluent with the curvature describing the distal contact area
Mandibular 2nd Premolar	Mesial outline is convex for a short distance near the cervical line.	The distal outline is more convex.
Mandibular 1st Molar	Somewhat concave at the cervical third then become convex	Straight above the cervical line and soon becomes convex forming the <i>distal contact area</i> at the middle third of crown.
Mandibular 2nd Molar	Convex	Convex
Mandibular 3rd Molar	Convex	Convex

Dental Anatomy

Shape	Buccal Lingual	Mesial Distal	Incisal Occlusal
Mandibular Central	Trapezoid	Triangular	Oval
Mandibular Lateral	Trapezoid	Triangular	Oval
Mandibular Canine	Trapezoid	Triangular	Oval
Mandibular 1 st Premolar	Trapezoid	Rhomboid	Diamond/Circular
Mandibular 2 nd Premolar	Trapezoid	Rhomboid	Square in 3-Cusp Circular in 2-Cusp
Mandibular 1 st Molar	Trapezoid	Rhomboid	Hexagonal with unequal sides
Mandibular 2 nd Molar	Trapezoid	Rhomboid	Rectangular
Mandibular 3 rd Molar	Trapezoid	Rhomboid	Rectangular Quadrilateral

Dental Anatomy

Shape	Buccal Lingual	Mesial Distal	Incisal Occlusal
Maxillary Central	Trapezoid	Triangular	Triangular
Maxillary Lateral	Trapezoid	Triangular	Oval
Maxillary Canine	Trapezoid	Triangular	Diamond
Maxillary 1 st Premolar	Trapezoid	Trapezoid	Hexagonal
Maxillary 2 nd Premolar	Trapezoid	Trapezoid	Oval
Maxillary 1 st Molar	Trapezoid	Trapezoid	Rhomboid
Maxillary 2 nd Molar	Trapezoid	Trapezoid	Rhomboid in 4-Cusp Heart in 3-Cusp
Maxillary 3 rd Molar	Trapezoid	Trapezoid	Heart

Dental Anatomy

	Number of Cusps	Notes
Maxillary 1 st Premolar	2	Buccal (Well-developed, bigger, placed distal to the center of crown, the only cusp that's placed distally in teeth(feature)) Lingual(Shorter, smaller and sharp)
Maxillary 2 nd Premolar	2	Buccal(Blunt cusp tip, less pointed than Max 1 st Premolar) Lingual (Slightly shorter than the buccal cusp)
Maxillary 1 st Molar	5	Mesiolingual (Largest cusp) Mesiobuccal (Wider buccal cusp, both buccal cusps of same length) Distobuccal (More pointed than Mesiobuccal cusp) Distolingual (Small, spheroidal) Cusp of Carabelli (Lingual to mesiolingual cusp)
Maxillary 2 nd Molar	4 or 3	Buccal Cusps: Mesiobuccal (Larger) Distobuccal (Much smaller and shorter) Lingual Cusps: Mesiolingual (Larger and sharper) Distolingual (Shorter and blunt and might be absent in 3-Cusps type)
Maxillary 3 rd Molar	3 or 4	Buccal cusps: Mesiobuccal Distobuccal Lingual Cusps: Mesiolingual (Occupying most of the lingual aspect) Distolingual (Very small or might be absent)
Mandibular 1 st Premolar	2	Buccal cusp (Sharp and pointed, placed mesial to the center of crown) Lingual cusp (Much shorter, non-functional, appears as a large cingulum, pointed)
Mandibular 2 nd Premolar	2-Cusp Type 3-Cusp Type	2-Cusp type: Buccal Cusp (Blunt, buccal to the vertical root axis) Lingual Cusp (In line with lingual outline of root) 3-Cusp type: Buccal Cusp Lingual Cusps: Mesiolingual(Larger) Distolingual(Smaller)

Dental Anatomy

Mandibular 1st Molar	5	Mesiolingual(Sharp) Distolingual (Sharp, same width as mesiolingual) Mesiobuccal(Blunt, largest) Distobuccal(Blunt) Distal(Smallest)
Mandibular 2nd Molar	4	Lingual cusps: Sharper, of equal size and at higher level than buccal cusps Mesiolingual and Distolingual Buccal cusps: Blunt, equal in width Mesiobuccal and Distobuccal
Mandibular 3rd Molar	4	Mesiolingual, Distolingual, Mesiobuccal and Distobuccal Distobuccal and Distolingual cusps are at lower level than mesial cusps

Dental Anatomy

Outline	Mesial	Distal
Maxillary Central	Straight	Slightly Convex
Maxillary Lateral	Slightly Convex	More Convex
Maxillary Canine	Slightly Convex till the contact area	Slightly Concave until the contact area
Maxillary 1st Premolar	Slightly Concave from the cervical line to the mesial contact area Both outlines become convex at contact areas	Nearly straight from the cervical line to the distal contact areas Both outlines become convex at contact areas
Maxillary 2nd Premolar	Slightly Convex	More convex
Maxillary 1st Molar	Little concave then it become convex	Convex from the cervical line to the tip of distobuccal cusp
Maxillary 2nd Molar	Nearly a straight line extending mesio-occlusally up to the mesial contact area where it becomes convex	Convex from the cervix to the point where it joins the occlusal outline
Maxillary 3rd Molar	Nearly Straight	Convex

Dental Anatomy

Contact Area	Mesial	Distal
Maxillary central	Incisal 3 rd	Junction of Middle and Incisal 3 rd
Maxillary Lateral	Junction of Middle and Incisal 3 rd	Middle 3 rd
Maxillary Canine	Junction of Middle and Incisal 3 rd	Middle 3 rd
Maxillary 1 st Premolar	Middle 3 rd	Occlusally placed than mesial
Maxillary 2 nd Premolar	Occlusal to the center	At same level
Maxillary 1 st Molar	Junction of occlusal and middle 3 rd	Middle 3 rd
Maxillary 2 nd Molar	Junction of occlusal and middle 3 rd	At middle 3 rd
Maxillary 3 rd Molar	Middle 3 rd	No contact

Dental Anatomy

Contact Area	Mesial	Distal
Mandibular Central	Incisal 3 rd	Same level(Incisal 3 rd)
Mandibular Lateral	Incisal 3 rd	Incisal 3 rd but more cervically
Mandibular Canine	Incisal 3 rd	Middle 3 rd
Mandibular 1 st Premolar	Occlusal to the center of the crown	Same level
Mandibular 2 nd Premolar	Middle 3 rd	Middle 3 rd
Mandibular 1 st Molar	Junction of occlusal and middle 3 rd	Distal contour of distal cusp
Mandibular 2 nd Molar	Center of crown	Center of crown
Mandibular 3 rd Molar	Occlusal to center of crown	



دکتور حار سلیمان